WHAT IS NETWORK?

A **network** is a group of two or more devices that are connected to each other to **communicate and exchange information**.
In a computer network, devices such as **computers, smartphones, printers, servers, and routers** can be connected using wired or wireless connections.
The main purpose of a network is to allow devices to **share data, resources, and services**.
Example:
When our laptop connects to a Wi-Fi router, it becomes part of a network. Our laptop can then communicate with the router, access the internet, share files with other devices, or connect to a network printer.
A network can be **small**, such as a network connecting computers within a home, or **very large**, such as the **Internet**, which connects networks across the world.

What is Topology?

In a computer network, many devices such as computers, servers, printers, switches, and routers need to be connected with each other for communication and sharing of information. The way these devices are **arranged and interconnected** in a network is known as **network topology**.
Network topology is an important concept in computer networking because the arrangement of devices affects **how data travels, how efficiently the network works, its cost, reliability, and how easily the network can be expanded**. Different networks use different topologies depending on their requirements.
For example, a small office may connect all its computers to one central switch, while a larger network may use a combination of different topologies.
There are two main ways of looking at topology:
1. Physical Topology:
It describes the **actual physical arrangement** of devices and cables.
For example, if every computer in an office is physically connected to one central switch, the physical topology is a **star topology**.
2. Logical Topology:
It describes **how data actually flows** between devices, regardless of their physical arrangement.
For example, devices may be physically arranged in one way, but data may travel between them according to a different logical structure.
Example
Consider a computer laboratory with **20 computers**. If all 20 computers are connected to a **central switch**, and the switch controls communication between them, the network follows a **Star Topology**.
For example:
Computer 1 →
Computer 2 →
Computer 3 → Switch → Internet/Server
Computer 4 →
Computer 5 →
Here, the switch acts as the central connection point. If Computer 1 wants to send data to Computer 4, the data is sent through the central switch.

BUS TOPOLOGY

A bus topology is a network configuration in which every device (node) is connected to a single central cable, known as the bus, backbone, or trunk line. All the computers, printers, and other devices on the network share this one communication line to send and receive data.

Data sent by any device travels along the bus in both directions until it reaches the intended destination. Terminators are placed at each end of the cable to absorb the signal and prevent it from bouncing back, which would cause interference (signal reflection). Because all devices share the same cable, only one device can transmit data at a time.

Bus topology was widely used in early Ethernet networks (10BASE2 and 10BASE5) and is considered one of the simplest and oldest network topologies.

Advantages of Bus Topology

- Easy to install and set up compared to other topologies, since it requires less cabling.
- Cost-effective — uses less cable than star or mesh topologies, reducing hardware cost.
- Well suited for small networks with a limited number of devices.
- Simple to extend by adding a new device along the existing cable (using a T-connector or additional segment).
- Failure of one workstation does not necessarily affect the rest of the network (unlike ring topology).
- Requires less cable length, which also means less time and effort for installation

Disadvantages of Bus Topology

- If the main backbone cable fails or breaks, the entire network goes down.
- Network performance degrades as more devices are added, because all devices share the same bandwidth.
- Difficult to identify the exact location of a fault or cable break (troubleshooting is time-consuming).
- Limited cable length and number of devices that can be connected on a single segment.
- Data collisions are more likely since only one device can transmit at a time.
- Not very secure — data packets travel past every device on the bus, so any node can potentially read the traffic.
- Requires proper termination at both ends; a missing or faulty terminator disrupts the whole network.

Simulation / Design in Cisco Packet Tracer

This section presents the bus topology network designed and simulated in Cisco Packet Tracer. The network consists of multiple end devices (PCs) connected to a single common backbone cable, representing the shared communication medium of a bus topology.

Network Diagram / Topology Layout

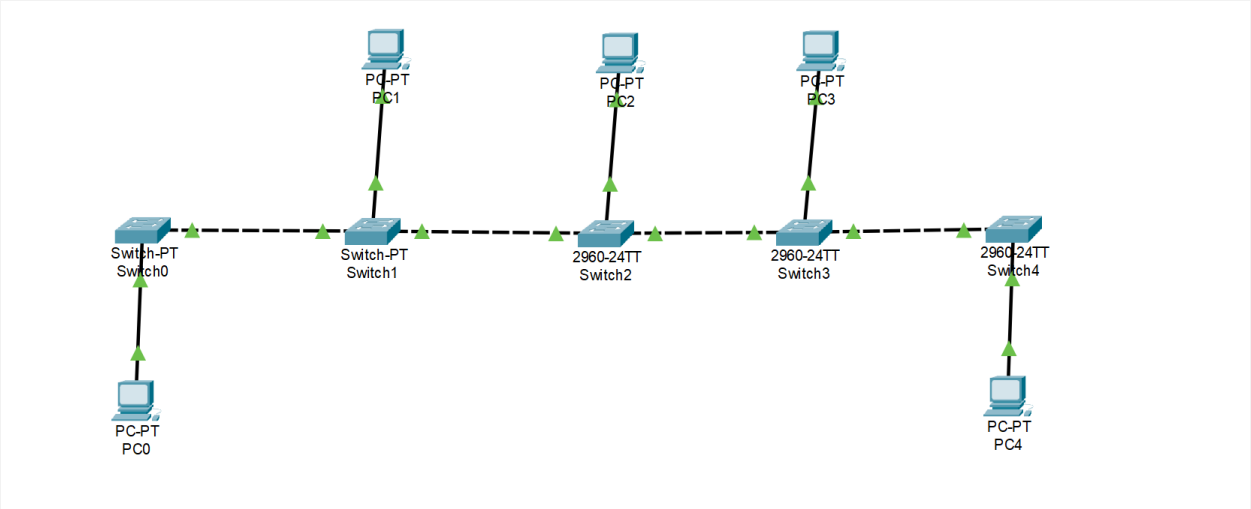


Figure 1: Bus topology layout in Cisco Packet Tracer

Device Configuration

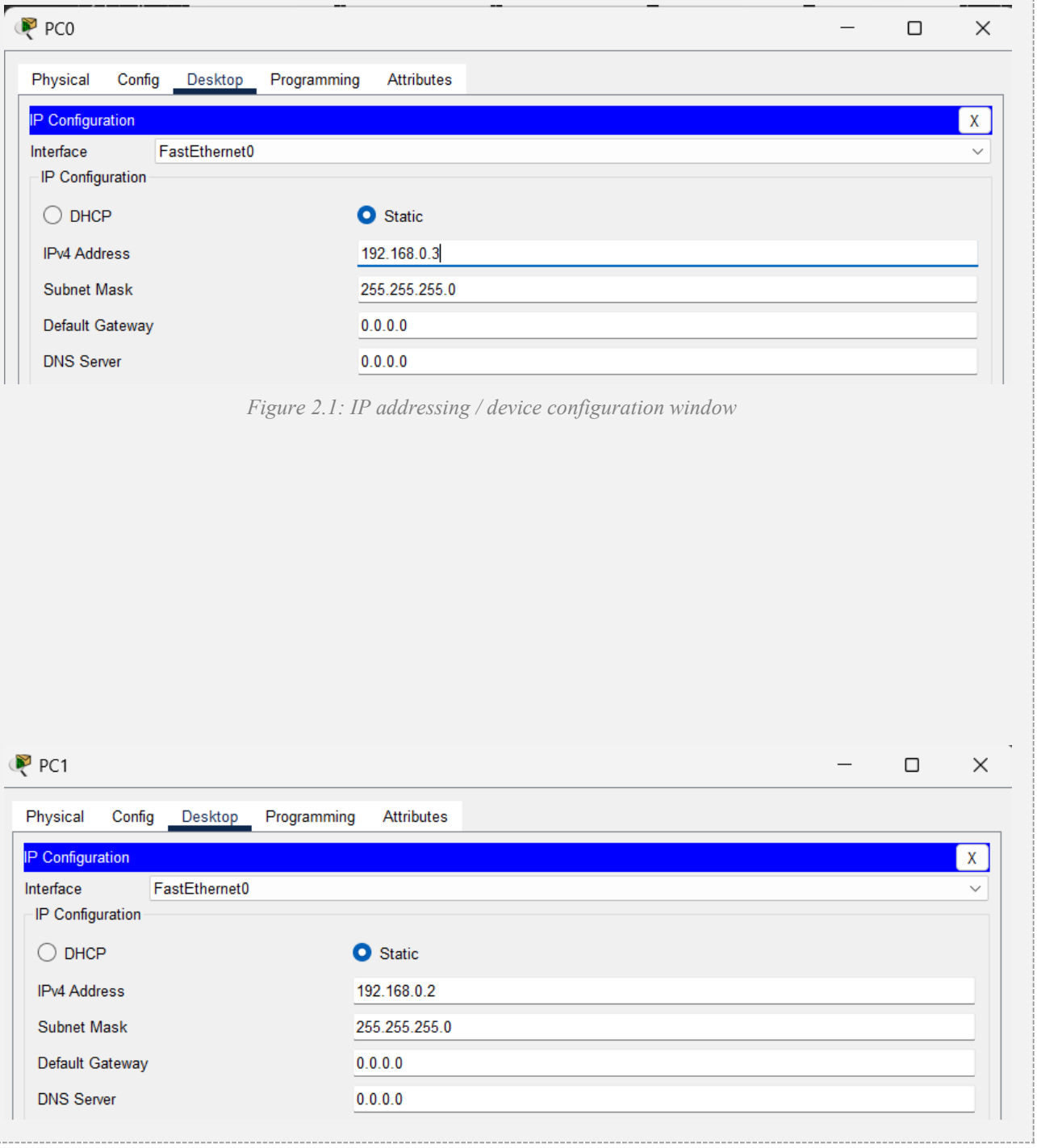


Figure 2.1: IP addressing / device configuration window

Figure 2.2: IP addressing / device configuration window

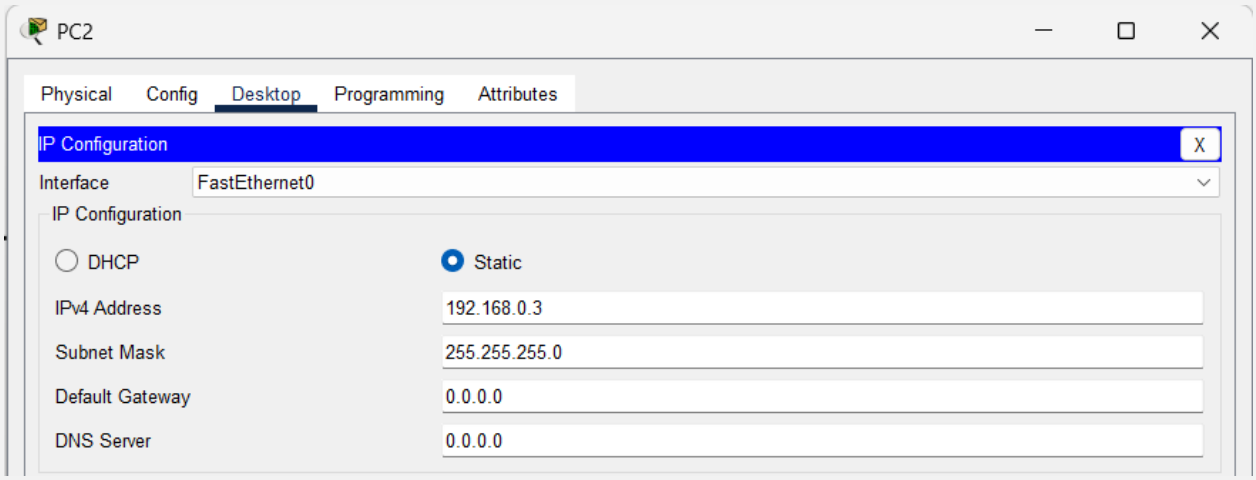


Figure 2.3: IP addressing / device configuration window

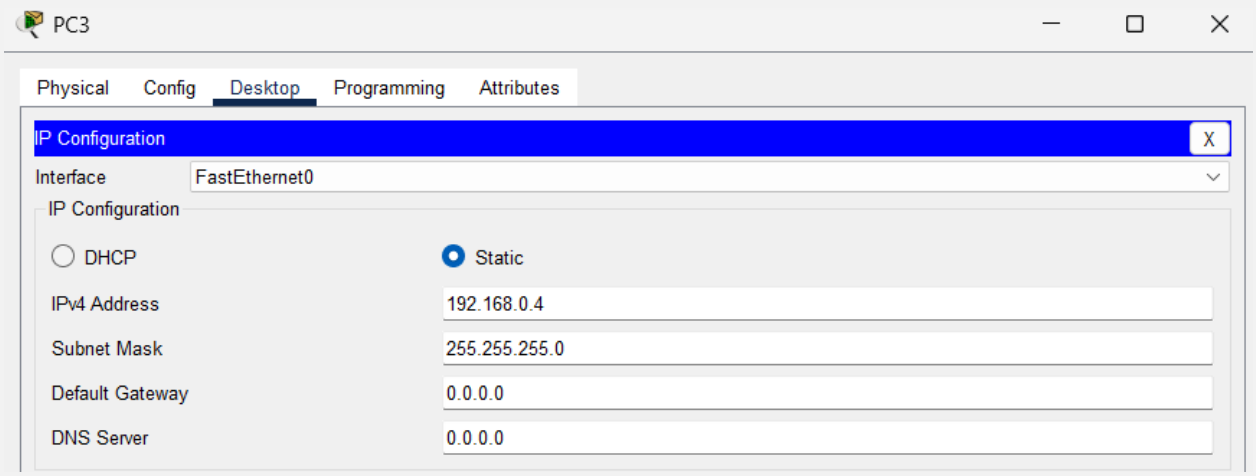


Figure 2.4: IP addressing / device configuration window

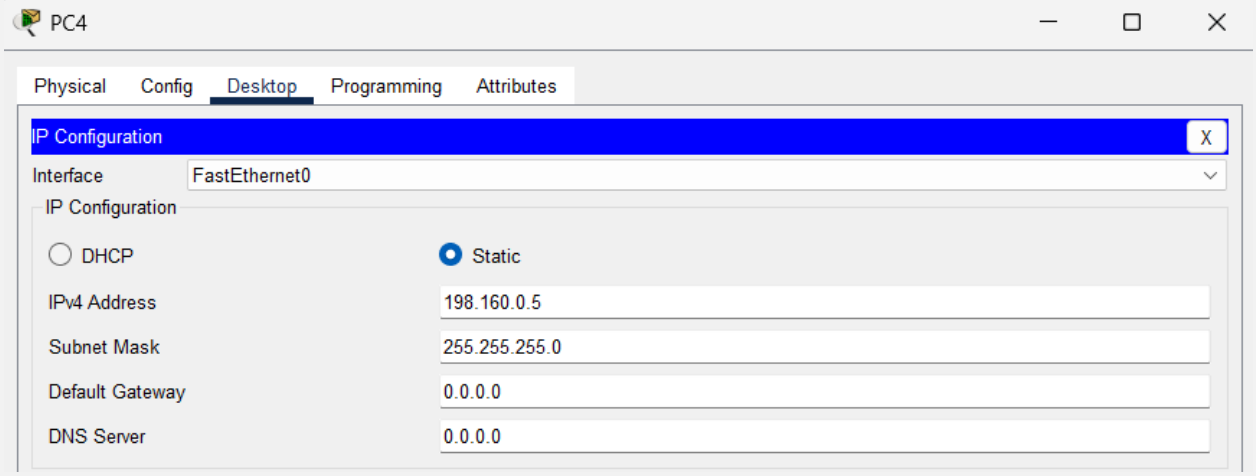


Figure 2.5: IP addressing / device configuration window

Connectivity Test (Ping / Simulation Result)

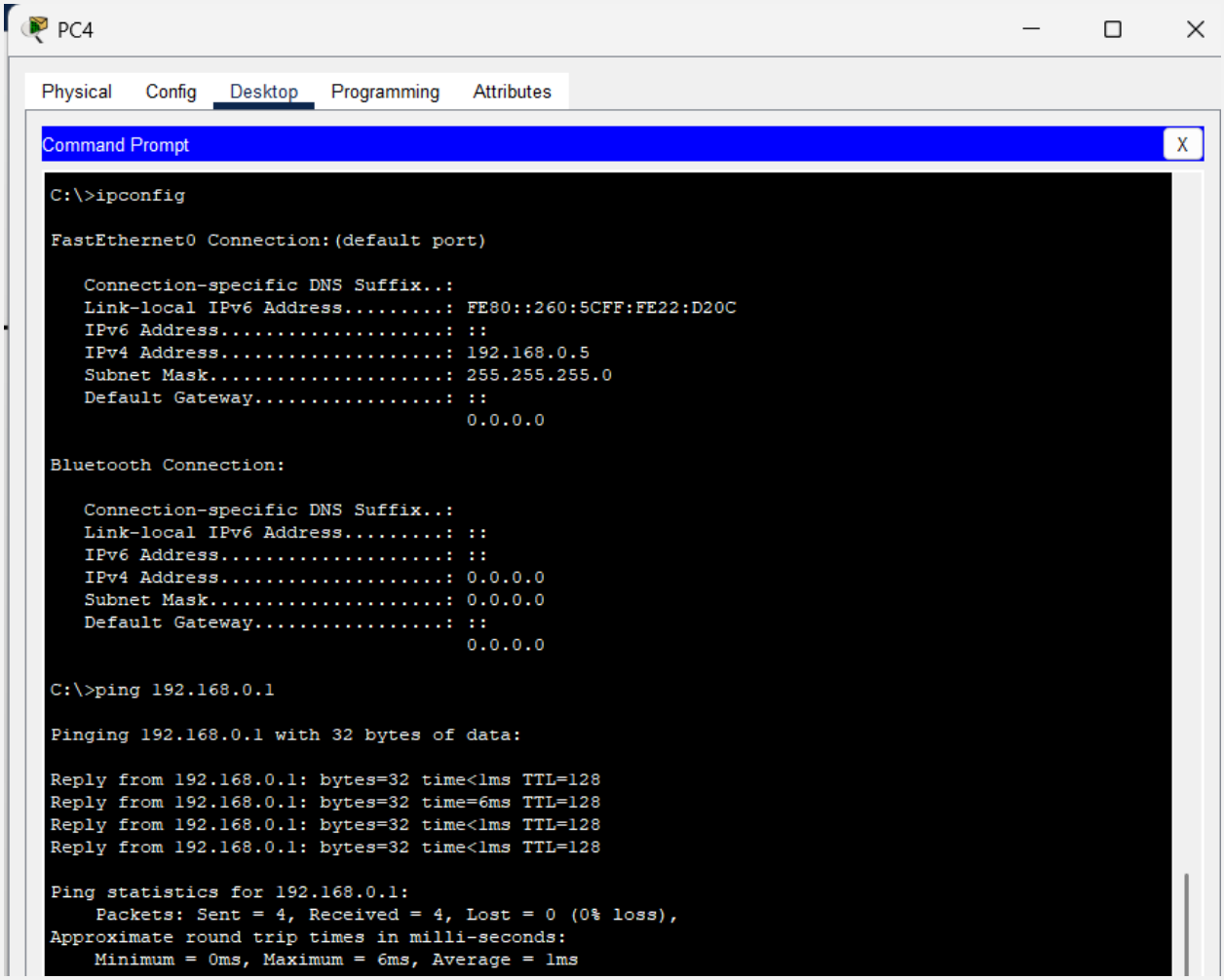


Figure 3: Successful ping / packet simulation result between PCs

Process of Designing the Network in Packet Tracer

I. Open Cisco Packet Tracer and create a new blank project to begin designing the network topology.

II. Select the required end devices, such as PCs, from the End Devices category and place them on the workspace. The PCs are arranged in a straight line to give the network the appearance of a bus topology.

III. Select a suitable connection from the Connections category to represent the shared communication medium. In a traditional bus topology, all devices use a common backbone cable, while in Packet Tracer, a suitable cable or hub arrangement can be used to represent this shared medium.

IV. Connect all the PCs to the same shared medium and make sure that each device has a proper physical connection with the network.

V. Configure the network settings of each PC by opening Desktop → IP Configuration. A unique IP address and a suitable subnet mask are assigned to every PC, with all the devices placed within the same network so that they can communicate with one another.

VI. After completing the connections and configurations, check the link indicators between the devices. Green indicators show that the physical connections are active and the devices are properly connected.

VII. The network can then be tested using Simulation Mode. The Add Simple PDU option is used to send a test packet from one PC to another, allowing the movement of the packet through the network to be observed.

VIII. Connectivity can also be checked through the Command Prompt of a PC by using the ping command with the IP address of another PC. For example, if the destination PC has the IP address 192.168.0.3, a ping can be sent to that address. A successful reply confirms that communication between the two PCs has been established.

IX. Screenshots of the completed topology, IP configurations, and successful connectivity test are captured as evidence of the completed network setup.

X. Finally, the completed Packet Tracer project is saved as a .pkt file, along with the required documentation and screenshots for submission. Open Cisco Packet Tracer and create a new blank project to begin designing the network topology.

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Conclusion

The bus topology is a simple and low-cost network design suitable for small networks, though it has notable drawbacks such as a single point of failure at the backbone cable and reduced performance as devices increase. Simulating the topology in Cisco Packet Tracer helps visualize how devices share a common communication medium and reinforces understanding of both its advantages and limitations.